

# goxpyriment — User Manual

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## goxpyriment User's Manual

This manual explains the key concepts of the library. It assumes you have read the [Getting Started](#) guide and want to understand the framework well enough to write experiments confidently.

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## 1. The Experiment Object

Every experiment revolves around a single `*control.Experiment` value. It is a façade that owns the SDL window, renderer, default font, audio device, keyboard and mouse handlers, and data file. You never create these separately.

```
exp := control.NewExperimentFromFlags("My Experiment", control.Black, control.White, 32)
defer exp.End()
```

`NewExperimentFromFlags` parses two optional command-line flags:

| Flag | Effect                                                                |
|------|-----------------------------------------------------------------------|
| -w   | Windowed mode: opens a 1024×768 window instead of going fullscreen    |
| -d N | Display ID: open on monitor N (-1 = primary display)                  |
| -s N | Set subject ID to N (integer); written to the data file automatically |

`defer exp.End()` must appear immediately after construction. It saves the data file, releases fonts, destroys the window, and shuts down SDL — even if the experiment panics or returns early.

### Key fields

```
exp.Screen      *apparatus.Screen      // window + renderer
exp.Keyboard    *apparatus.Keyboard    // keyboard input
```

```
exp.Mouse          *apparatus.Mouse          // mouse input
exp.AudioDevice    sdl.AudioDeviceID         // SDL audio device for Sound/Tone
exp.Audio          *control.AudioManager     // high-level audio helpers
exp.Data           *results.DataFile         // output data file
exp.Design         *design.Experiment        // block/trial structure
exp.SubjectID      int
exp.DefaultFont    *ttf.Font
exp.Info           map[string]string        // values collected by GetParticipantInfo, if use
```

## 2. Collecting Participant Information

Before the experiment window opens, you can display a graphical setup dialog that lets the experimenter fill in participant demographics, monitor characteristics, and display mode. The dialog returns the collected values as a `map[string]string`.

```
fields := append(control.StandardFields, control.FullscreenField)
info, err := control.GetParticipantInfo("My Experiment", fields)
if err != nil {
    log.Fatalf("setup cancelled: %v", err)
}
```

Call `GetParticipantInfo` **before** `NewExperiment` or `NewExperimentFromFlags`. It initializes SDL internally, shows the window, and shuts SDL down again before returning, so the subsequent experiment initialization starts from a clean state.

### Pre-built field sets

| Variable                               | Fields included                                                                                   |
|----------------------------------------|---------------------------------------------------------------------------------------------------|
| <code>control.ParticipantFields</code> | <code>subject_id</code> , <code>age</code> , <code>gender</code> , <code>handedness</code>        |
| <code>control.MonitorFields</code>     | <code>screen_width_cm</code> , <code>viewing_distance_cm</code> ,<br><code>refresh_rate_hz</code> |
| <code>control.StandardFields</code>    | <code>ParticipantFields</code> + <code>MonitorFields</code><br>combined                           |
| <code>control.FullscreenField</code>   | Checkbox: <code>fullscreen</code> ("true" / "false")                                              |

### Defining custom fields

```
fields := []control.InfoField{
    {Name: "subject_id", Label: "Subject ID", Default: ""},
    {Name: "session", Label: "Session (1/2/3)", Default: "1"},
    {Name: "room", Label: "Testing room", Default: "Lab A"},
    {Name: "fullscreen", Label: "Fullscreen mode", Default: "true",
        Type: control.FieldCheckbox},
}
info, err := control.GetParticipantInfo("My Experiment", fields)
```

Fields of type `FieldText` (the default) render as text input boxes. Fields of type `FieldCheckbox` render as tick-boxes; their value is always `"true"` or `"false"`.

## Dialog interaction

| Action                | Effect                                         |
|-----------------------|------------------------------------------------|
| Click a field         | Focus it                                       |
| Type                  | Append text to the focused field               |
| Backspace             | Delete last character                          |
| Tab / Shift-Tab       | Move focus to next / previous text field       |
| Enter                 | Confirm (same as clicking OK)                  |
| Escape / close window | Cancel — <code>ErrCancelled</code> is returned |

## Session persistence

All values except `subject_id` are saved to `~/.cache/goxpyriment/last_session.json` when the experimenter confirms. They are pre-filled on the next run. `subject_id` is always reset — the experimenter must enter it fresh each session.

## Using the results

```
info, err := control.GetParticipantInfo("My Experiment", fields)
if err != nil {
    log.Fatalf("setup cancelled: %v", err)
}

// Use the fullscreen checkbox to choose the window mode
fullscreen := info["fullscreen"] == "true"
width, height := 0, 0
if !fullscreen {
    width, height = 1024, 768
}

exp := control.NewExperiment("My Experiment", width, height, fullscreen,
    control.Black, control.White, 32)

// Set Info and SubjectID BEFORE Initialize so they are written to the .csv header
exp.SubjectID, _ = strconv.Atoi(info["subject_id"])
exp.Info = info

if err := exp.Initialize(); err != nil {
    log.Fatalf("failed to initialize: %v", err)
}
defer exp.End()
```

When `exp.Info` is non-nil at the time `Initialize()` is called, the collected key/value pairs are automatically written as a `--PARTICIPANT INFO` block in the `.csv` metadata header — no extra call is required.

**Note:** When using `GetParticipantInfo` you will typically call the lower-level `NewExperiment + Initialize()` instead of `NewExperimentFromFlags`, so you can pass the fullscreen/windowed choice from the dialog directly.

---

### 3. The Run Loop and Error Handling

#### The `exp.Run` wrapper

All experiment logic runs inside a callback passed to `exp.Run`:

```
err := exp.Run(func() error {
    // your experiment here
    return control.EndLoop
})
if err != nil && !control.IsEndLoop(err) {
    log.Fatalf("experiment error: %v", err)
}
```

`exp.Run` does two important things:

1. **Ensures everything runs on the SDL main thread.** SDL requires that all rendering and event calls happen on the thread that created the window. `exp.Run` guarantees this.
2. **Installs a panic/recover guard.** When the participant presses ESC, the library immediately aborts the trial loop by calling `panic` with an internal sentinel value. `exp.Run` catches this panic, converts it back to an error, and returns it cleanly. Your data is saved.

#### Returning from the callback

| Return value                 | Meaning                                                |
|------------------------------|--------------------------------------------------------|
| <code>control.EndLoop</code> | Normal exit; experiment is done                        |
| <code>nil</code>             | Continue — call the callback again on the next frame   |
| any other error              | Propagated as the return value of <code>exp.Run</code> |

In a typical experiment you run the full trial loop inside a single callback invocation and return `control.EndLoop` at the end. The `nil` / “keep looping” pattern is used for frame-by-frame animation; in most cases you will never return `nil`.

#### You don’t need to check every error

Because pressing ESC triggers the panic/recover mechanism, any call that would have returned an error (e.g., `exp.Show`, `exp.Wait`, `exp.Keyboard.WaitKey`) will instead abort

the loop immediately. The error never reaches your code.

This means the following two styles are both correct:

```
// Style A – check errors explicitly (safer for library/production code)
if err := exp.Show(fixation); err != nil {
    return err
}
if err := exp.Wait(500); err != nil {
    return err
}

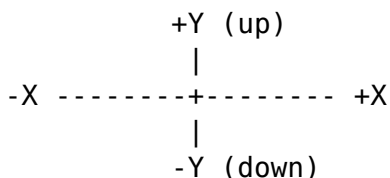
// Style B – omit error checks inside exp.Run (fine for experiment scripts)
exp.ShowTimed(fixation, 500)
```

Style B works because if something goes wrong (ESC pressed, window closed), the panic/recover mechanism unwinds the call stack before your code can observe an error. Use Style A if you have cleanup that must run on abort; otherwise Style B keeps experiment scripts readable.

---

## 4. The Coordinate System

All stimulus positions use a **center-origin** system:



- (0, 0) is the screen center.
- Positive Y is **up** (opposite to SDL's default, which is top-down).
- Units are pixels at the logical resolution.

```
// Center of screen
stimuli.NewTextLine("Hello", 0, 0, control.White)

// 200 px to the right
stimuli.NewCircle(30, control.Red).SetPosition(control.Point(200, 0))

// Bottom-left area
stimuli.NewTextLine("Score", -400, -250, control.White)
```

The conversion to SDL coordinates (top-left origin) is handled internally by `screen.CenterToSDL(x, y)`. You never call this yourself unless you are writing a custom stimulus type — in which case `screen.CenteredRect(pos, w, h)` returns the SDL destination rectangle for a `w`×`h` texture centered at `pos`, factoring out the convert-and-offset step.

## Logical size

If you call `exp.SetLogicalSize(width, height)`, the coordinate system scales to that virtual resolution regardless of the actual window or screen size. This is useful for making experiments resolution-independent:

```
if err := exp.SetLogicalSize(1920, 1080); err != nil {
    log.Printf("warning: %v", err)
}
// Now (960, 540) is the center-right area, regardless of actual screen size.
```

## Display scaling (HiDPI / fractional scaling)

**Recommendation: set your OS display scaling to 100% before running experiments.**

Most desktop environments (GNOME, KDE, Windows, macOS) allow the user to scale the entire UI — e.g. 125%, 133%, 150% — to make text and icons larger on high-density screens. Goxpyriment does not currently compensate for fractional OS-level scaling. Running with a scale factor other than 100% can cause:

- stimulus positions and sizes to be wrong (coordinates are in logical pixels, but the OS maps them to physical pixels at the scale factor),
- video playback and canvas rendering to appear cropped or misaligned,
- the window to not fill the screen correctly in fullscreen mode.

To disable scaling before running an experiment:

- **GNOME (Ubuntu):** *Settings* → *Displays* → *Scale* → 100%
- **KDE:** *System Settings* → *Display and Monitor* → *Scale* → 100%
- **Windows:** *Settings* → *Display* → *Scale* → 100%
- **macOS:** *System Settings* → *Displays* → *Resolution* → *Default* (or choose a non-HiDPI mode)

You can restore your preferred scaling after the session.

---

## 5. The Rendering Model

SDL uses a **double-buffered** rendering model. There is an off-screen backbuffer where you draw, and the visible display. You draw to the backbuffer; calling `screen.Update()` (also called a “flip”) presents it to the screen, synchronized to the vertical retrace (VSYNC).

The three-step cycle for showing one stimulus is:

```
exp.Screen.Clear()           // fill backbuffer with background color
myStim.Draw(exp.Screen)      // draw stimulus onto backbuffer
exp.Screen.Update()          // present to display (VSYNC-blocks)
```

`exp.Show(stim)` does all three in one call and is the right choice for single-stimulus presentations. For cases where you need to draw multiple stimuli simultaneously —

so they appear on screen at the same time — call each Draw separately before the single Update:

**Triple/mailbox buffering.** On some systems (NVIDIA + compositor, Wayland) the system may present in a non-blocking mode — mailbox (“triple buffering”), where a newer frame replaces the pending one and `SDL_RenderPresent` returns immediately, or via a compositor that accepts your buffer rather than blocking on scanout. The display stays tear-free (it still updates only at VSYNC), but the *call* no longer paces one frame per call: it does **not** block. In a per-frame loop, the duration loop then completes without consuming wall-clock time, so the stimulus is replaced too early — its on-screen duration is unpredictably short (often, but not always, zero frames). Wayland is especially affected because presentation is driven by frame callbacks rather than a blocking swap, and reliable FIFO vsync requires recent compositor protocols. Use `screen.PacedFlip()` instead of `screen.Update()` in any loop that presents a stimulus for multiple frames: it calls `Update()` and, if the call returned before the expected frame boundary, busy-waits until the correct time. The pacing state is stored on `Screen` automatically; no extra variables are needed.

```
// Show fixation cross and stimulus simultaneously
exp.Screen.Clear()
fixation.Draw(exp.Screen)
targetCircle.Draw(exp.Screen)
exp.Screen.Update()
```

## The blank screen

`exp.Blank(ms)` clears the screen, flips it (showing a blank), and then waits:

```
exp.Blank(1000) // 1-second blank inter-trial interval
```

This is equivalent to `exp.Screen.Clear() + exp.Screen.Update() + exp.Wait(ms)`.

## Never draw outside `exp.Run`

All rendering must happen inside the `exp.Run` callback — equivalently, on the SDL main thread. Drawing from a goroutine will silently do nothing or crash.

---

## 6. Timing Architecture

### Frame cadence

`screen.Update()` blocks until the display’s vertical retrace (VSYNC). On a 60 Hz monitor this is ~16.67 ms; on a 120 Hz monitor ~8.33 ms. This is the fundamental clock of the visual display: every stimulus change is aligned to a frame boundary automatically.



On some systems (NVIDIA proprietary driver + compositor, Wayland) the system may select a non-blocking presentation mode (mailbox/“triple buffering”) or route the swap through a compositor, so `Update()` returns before the frame is scanned out. The display is still tear-free, but you cannot rely on the call to block for one frame, and a per-frame loop then runs too fast — shortening or dropping multi-frame stimuli. (This is not VSYNC being disabled: a blocking mode such as FIFO exists; the system just isn’t guaranteed to use it. Wayland paces via frame callbacks rather than a blocking swap.) Use `screen.PacedFlip()` as a drop-in for `screen.Update()` in any tight frame loop — it adds a busy-wait when needed and is a no-op on systems with correct VSYNC blocking.

To know your frame duration at runtime:

```
frameDur := exp.Screen.FrameDuration() // e.g., 16.666ms on a 60 Hz display
fmt.Printf("Frame: %.2f ms\n", frameDur.Seconds()*1000)
```

### **exp.Wait vs clock.Wait**

| Function                    | Pumps SDL events? | Detects ESC? | Use when                                         |
|-----------------------------|-------------------|--------------|--------------------------------------------------|
| <code>exp.Wait(ms)</code>   | Yes               | Yes          | Inside <code>exp.Run</code> ,<br>between stimuli |
| <code>clock.Wait(ms)</code> | No                | No           | Short busy-waits,<br>inside animation<br>loops   |

Always use `exp.Wait` for inter-trial and inter-stimulus intervals — it keeps the OS responsive and responds to ESC. Use `clock.Wait` only inside VSYNC-locked loops that handle events themselves (streams, animation loops).

### **Sub-millisecond timing is not guaranteed for exp.Wait**

`exp.Wait` sleeps in 1 ms increments. For coarse delays (ISIs, fixation durations) this is perfectly fine. For frame-accurate stimulus timing — e.g., showing a stimulus for exactly 2 frames — use the stream functions described in [Section 10](#).

### **Nanosecond event timestamps**

SDL3 timestamps every input event **at hardware-interrupt time**, before any application code runs. The timestamp is stored with the event in the SDL event queue, where it remains untouched until your code reads it.

This has a crucial consequence: **it does not matter when you call `GetKeyEventTS`**. Any keypress that occurred — even during `exp.Wait`, between two `ShowTS` calls, or while other computation was running — is waiting in the queue with its exact hardware timestamp. Nothing is lost. This is fundamentally different from a polling-based approach, where a keypress can only be measured from the moment the polling function is called, and any code running before that call inflates the apparent RT.

`exp.ShowTS(stim)` records the SDL nanosecond time of the VSYNC flip. `GetKeyEventTS` retrieves the event’s own hardware timestamp. Subtracting the two gives hardware-precision RT regardless of how much code ran in between:

```
onset, _ := exp.ShowTS(stim1) // record flip timestamp
exp.Wait(500)                // keypress here is NOT lost
exp.ShowTS(stim2)            // keypress here is also NOT lost
key, keyTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, -1)
rtToStim1 := int64(keyTS - onset) // correct even if pressed during Wait
```

This is especially useful when you need RT relative to a specific stimulus in a multi-stimulus sequence — something that `WaitKeysRT` cannot express directly, since it measures from the call site rather than from a recorded onset.

## Two clocks: the Go clock and the SDL clock

`goxpyriment` exposes time through **two independent clocks with different zero points**. A value read from one must never be subtracted from a value read from the other — the difference would silently include the unknown, machine-dependent offset between their origins.

| Clock                     | Read it via                                                                                                                                                                                                                                                                           | Zero point                                                             | Use it for                                                                                                                                                                                                  |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Go monotonic clock</b> | <code>clock.GetTime()</code> ,<br><code>clock.GetTimeNS()</code> ,<br><code>Clock.Now()</code> ,<br><code>Clock.NowMillis()</code> ,<br><code>Clock.NowNanos()</code>                                                                                                                 | First use of the<br>clock package, or<br><code>clock.NewClock()</code> | <i>Scheduling and logging in your own code:</i><br>inter-trial and inter-stimulus intervals, drift-free trial pacing ( <code>SleepUntil</code> ), and “time since start of block” columns in your data file |
| <b>SDL event clock</b>    | <code>exp.ShowTS()</code> ,<br><code>Screen.FlipTS()</code> ,<br><code>Screen.PacedFlipTS()</code> ,<br>Key-board.<br><code>GetKeyEventTS()</code> ,<br><code>WaitAnyEventTS()</code> ,<br>and the <code>Timestamp</code> field of any SDL event (all on <code>sdl.TicksNS()</code> ) | SDL initialisation                                                     | <b>Reaction-time measurement:</b><br>whenever you subtract a stimulus onset from a response, <i>both</i> values must come from this clock                                                                   |

**The rule.** Measure reaction times entirely on the SDL clock — subtract a `FlipTS/ShowTS` onset from a `GetKeyEventTS` response. Use the Go clock only for things you *schedule* rather than *measure*: ISIs, fixation durations, block pacing,

and log timestamps. Crossing the two — e.g. `keyTS - clock.GetTimeNS()` — is meaningless.

**Why two clocks?** The SDL clock is the one SDL stamps onto hardware input events at interrupt time (see the previous subsection), so it is the only clock that can measure RT without inflation from intervening code. The Go clock needs no SDL context, is what `exp.Wait`, `clock.Wait`, and `Clock.SleepUntil` are built on, and is the natural choice for scheduling and for log timestamps that just need to be internally consistent.

The two roles coexist cleanly in a single trial loop — Go clock for pacing and logging, SDL clock for the RT:

```
c := clock.NewClock() // Go clock: pacing + logging
for i, trial := range block.Trials {
    c.SleepUntil(time.Duration(i) * 2 * time.Second) // Go clock: drift-free onset ca
    onset, _ := exp.ShowTS(trial.Stimulus) // SDL clock: stimulus onset
    key, keyTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, -1)
    rtMs := float64(keyTS-onset) / 1e6 // SDL - SDL ✓ valid RT
    exp.Data.Add(trial.Label, key, rtMs, c.NowMillis()) // log time on the Go clock
}
```

Here `keyTS - onset` stays within the SDL clock (a real RT), while `c.SleepUntil` and `c.NowMillis()` stay within the Go clock (scheduling and a log column). The two are never subtracted from each other.

## Disabling garbage collection

Go's garbage collector can pause execution for several milliseconds at unpredictable times. The stream and animation functions disable it during the critical loop:

```
old := debug.SetGCPercent(-1)
defer debug.SetGCPercent(old)
```

You do not need to do this yourself for ordinary trial loops; only for high-speed RSVP or animation. The stream and animation functions handle it automatically.

## Choosing a timing approach

Use this decision guide before committing to a timing strategy.

| Situation                                           | Recommended approach                                                                                                        |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Coarse ISIs and fixation durations ( $\geq 100$ ms) | <code>exp.Wait(ms)</code> — adequate precision, keeps the event loop alive                                                  |
| Show stimulus + timed hold (fixation, cue, mask)    | <code>exp.ShowTimed(stim, ms)</code> — Show + Wait in one call                                                              |
| Single-stimulus trial, RT relative to onset         | <code>key, rt, err := exp.ShowAndGetRT(stim, keys, timeout)</code> — hardware-precise RT, clears stale events automatically |

| Situation                                                                | Recommended approach                                                                                                             |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Stimulus duration measured in frames (e.g. 2-frame mask)                 | Per-frame loop with <code>screen.PacedFlip()</code> (safe on all drivers), or <code>PresentStreamOfImages</code> (Section 10)    |
| Multi-stimulus sequence, RT relative to a specific stimulus              | <code>exp.ShowTS(stim) + exp.Keyboard.GetKeyEventTS(...)</code> — nanosecond timestamps on the same SDL clock; subtract directly |
| RSVP or rapid animation (frame-accurate presentation of many items)      | <code>PresentStreamOfImages / PresentStreamOfText</code> (Section 10) — GC disabled, VSYNC-locked, full timing log               |
| EEG/MEG synchronisation required                                         | Add a TTL pulse via triggers immediately after <code>exp.ShowTS</code> (Section 15)                                              |
| Stimulus duration not a multiple of the frame period (e.g. 10 ms, 17 ms) | Variable Refresh Rate mode — see below                                                                                           |

**OS considerations.** Presentation latency depends heavily on whether the display compositor is active — see the next section. For EEG work, always verify onset timing with a photodiode regardless of OS.

**When in doubt, use `ShowAndGetRT`.** It clears stale events, records the VSYNC flip timestamp, waits for the key with hardware-precision timing, and returns (`key`, `rtMs`, `error`) — the canonical single-stimulus RT call. Use raw `ShowTS` + `GetKeyEventTS` only when composing multiple stimuli before a single response.

**`ShowTS` vs `FlipTS` vs `PacedFlipTS`.** `exp.ShowTS(stim)` is the standard high-level call: it clears the screen, draws the stimulus, flips, and returns the flip timestamp — one line for the common case. `exp.Screen.FlipTS()` is the lower-level primitive that only flips and timestamps, leaving clearing and drawing to you. `exp.Screen.PacedFlipTS()` is the same as `FlipTS` but adds the triple-buffering guard (busy-wait if the driver returned early); use it for the onset frame of a multi-frame stimulus in a per-frame loop. Use plain `FlipTS` only in VRR mode (where VSYNC is disabled by design) or when you have confirmed that the driver blocks correctly.

## Display compositor bypass

A compositing window manager blends every application frame with other on-screen elements before sending the result to the display. This adds latency and jitter that can be significant for millisecond-accurate stimulus presentation. How much overhead you incur — and whether you can avoid it — depends on the operating system.

**Linux / X11 (fullscreen).** SDL3 automatically sets the `_NET_WM_BYPASS_COMPOSITOR` window property when the application enters fullscreen mode. Compliant compositing WMs (KWin, Mutter, Compton) respond by *unredirecting* the fullscreen window: the application’s framebuffer is routed directly to the display scan-out pipeline without compositing. Presentation latency drops below 1 ms — comparable to a compositor-

free session. Nothing special needs to be done in goxpyriment; running fullscreen (omit `-w`) is sufficient.

**Linux / Wayland (fullscreen).** The Wayland compositor is always the intermediary for frame delivery, but modern compositors (KWin 5.21+, GNOME Mutter 44+) support *direct scan-out* for fullscreen applications, which routes the framebuffer to the display hardware with near-zero overhead. In practice the latency is similar to X11 bypass, though this depends on the compositor version and GPU driver.

**Linux / KMS-DRM (no display server).** The lowest-latency configuration on Linux is to run without any display server at all, from a virtual terminal (Ctrl+Alt+F2). Setting the environment variable `SDL_VIDEODRIVER=kmsdrm` before launching the experiment directs SDL3 to use the Linux kernel's KMS/DRM subsystem directly, bypassing both X11 and Wayland entirely:

```
SDL_VIDEODRIVER=kmsdrm go run myexperiment/main.go
```

This is the recommended configuration for the most demanding timing requirements, such as single-frame subliminal stimulation or high-frequency RSVP.

**Windows (fullscreen).** SDL3's fullscreen mode creates an exclusive fullscreen surface that bypasses the Desktop Window Manager (DWM), yielding frame-interval jitter typically below 0.3 ms (one standard deviation), comparable to Linux.

**macOS.** Apple's Metal pipeline always delivers frames through the WindowServer compositor, and no third-party application can bypass it — even in fullscreen.

These platform differences can be measured empirically with the display and frames sub-tests of the bundled Timing-Tests suite (see [TimingTests.md](#)).

## Variable Refresh Rate (VRR) — arbitrary stimulus durations

On a standard 60 Hz monitor, every stimulus duration must be a multiple of 16.67 ms. You cannot present a stimulus for 10 ms or 20 ms — the display simply cannot change in between frame boundaries. Variable Refresh Rate technology (AMD FreeSync, NVIDIA G-Sync, VESA Adaptive-Sync) removes this constraint: the panel holds each frame for exactly as long as the software asks, refreshing only when the next `Present()` call arrives.

**Enabling VRR in goxpyriment** goxpyriment opens the renderer with VSYNC enabled by default, which locks every flip to a frame boundary. To enter VRR mode, disable VSYNC before the timing-critical section and restore it afterwards:

```
// Switch to VRR mode – Present() now returns immediately.
if err := exp.Screen.SetVSync(0); err != nil {
    log.Fatalf("VRR not supported: %v", err)
}
defer exp.Screen.SetVSync(1) // always restore

// Pre-load stimulus textures before disabling GC.
stimuli.PreloadVisualOnScreen(exp.Screen, myStim)
```

```

old := debug.SetGCPercent(-1)
defer debug.SetGCPercent(old)

// Draw stimulus and present (returns immediately with VSync=0).
exp.Screen.Renderer.SetDrawColor(0, 0, 0, 255)
exp.Screen.Clear()
myStim.Draw(exp.Screen)
onsetNS, _ := exp.Screen.FlipTS()

// Hold for exactly targetDur using a busy-wait (sub-ms precision).
deadline := time.Now().Add(targetDur)
for time.Now().Before(deadline) { /* spin */ }

// Present blank – the display switches immediately on VRR.
exp.Screen.Renderer.SetDrawColor(0, 0, 0, 255)
exp.Screen.Clear()
offsetNS, _ := exp.Screen.FlipTS()

actualDurMs := float64(offsetNS-onsetNS) / 1e6

```

onsetNS and offsetNS are both `sdl.TicksNS()` timestamps on the SDL nanosecond clock, captured right after each `Present()` returns. Their difference measures the software-controlled stimulus duration. On a VRR display this equals the actual on-screen duration plus a small, constant hardware pipeline latency (GPU scan-out + panel response, typically 1–5 ms) that can be measured independently with the frames test and a photodiode.

**VRR range and self-diagnosis** Every VRR panel has a supported refresh-rate range (e.g. 48–144 Hz on a typical gaming monitor). Requesting a duration shorter than  $1000/\text{max\_Hz}$  ms or longer than  $1000/\text{min\_Hz}$  ms takes the panel outside its VRR window; the display then falls back to fixed-rate behaviour and durations are again quantised to frame multiples.

You can characterise your panel’s VRR window empirically with the Timing-Tests suite:

```
go run tests/Timing-Tests/main.go -test vrr -vrr-max-ms 50 -cycles 5 -w
```

This sweeps target durations from 1 ms to 50 ms in 1 ms steps and reports the actual vs. target duration at each step. Duration errors below 0.5 ms across the sweep confirm that VRR is working; large periodic errors confirm that VRR is absent or the requested duration is outside the supported range. See [TimingTests.md — VRR section](#) for detailed interpretation, including how to read the VRR boundary from the output and how to enable FreeSync on Linux.

## Requirements

- A VRR-capable monitor (FreeSync, G-Sync Compatible, or Adaptive-Sync).

- VRR enabled in the OS display settings (Linux: DRM/KMS or compositor; Windows: Display Settings → Advanced display → Variable refresh rate).
- The application does not need any special SDL hints; `SetVSync(0)` is sufficient.

**Note on tearing** On a monitor without VRR, `SetVSync(0)` disables the frame-boundary synchronisation entirely and the GPU writes to the framebuffer while the display is reading it, producing a horizontal tear line. On a VRR monitor this does not happen: the display waits for the GPU's signal before starting each new refresh. If your VRR test shows tearing artefacts, VRR is either not enabled in your OS or not supported by your hardware.

---

## 7. Input Handling

### Keyboard

The `exp.Keyboard` object provides blocking and non-blocking access:

```
// Block until any key – returns the keycode
key, err := exp.Keyboard.Wait()

// Block until a specific key
err := exp.Keyboard.WaitKey(control.K_SPACE)

// Block until one of several keys, with optional timeout (-1 = no timeout)
// Returns 0 on timeout, sdl.EndLoop on ESC/quit
key, err := exp.Keyboard.WaitKeys([]control.Keycode{control.K_F, control.K_J}, 3000)

// Same, but also returns reaction time in milliseconds from the call site
key, rt, err := exp.Keyboard.WaitKeysRT([]control.Keycode{control.K_F, control.K_J}, 3000)

// Hardware-precision version: returns the SDL event's nanosecond timestamp
key, eventTS, err := exp.Keyboard.GetKeyEventTS([]control.Keycode{control.K_F, control.K_J})

// Non-blocking poll – returns 0 if nothing pressed
key, err := exp.Keyboard.Check()

// Query whether a key is physically held down right now (no event queue involvement)
held := exp.Keyboard.IsPressed(control.K_SPACE)

// Wait for KEY_UP on a specific key; returns its SDL hardware timestamp (nanoseconds)
upTS, err := exp.Keyboard.WaitKeyReleaseTS(key, -1)

// Drain all pending key events (use before a new trial to discard stale presses)
exp.Keyboard.Clear()
```

**GetKeyEventTS “get” semantics.** Unlike the `Wait*` functions, `GetKeyEventTS` reads from the SDL event queue and returns *immediately* if a matching event is already



there — it only blocks when the queue has no matching event. This means a keypress that happened during `exp.Wait` or any other intervening code is consumed instantly on the next `GetKeyEventTS` call, with its original hardware timestamp intact.

**When to call `Clear()`.** Despite living on `exp.Keyboard`, `Clear()` drains the **entire** SDL event queue — keyboard, mouse, gamepad, and everything else — because SDL uses a single shared queue. Call it at the start of a trial (before `ShowTS`) to discard stale events from any device. Do **not** call it after `ShowTS`: the participant may have already responded, and `Clear()` would silently discard that event before `GetKeyEventTS` or `GetPressEventTS` ever sees it.

### **WaitKeysRT vs GetKeyEventTS:**

`WaitKeysRT` measures elapsed time from the moment the call is made (using `sdl.Ticks()`, millisecond granularity). In a single-stimulus trial this is a good approximation of RT from stimulus onset, but it accumulates any delay between `Show()` returning and `WaitKeysRT` being called.

`GetKeyEventTS` returns the SDL3 event's own `Timestamp` field — the nanosecond time at which the hardware key-down event was generated. Combined with `exp.ShowTS(stim)`, which records the nanosecond time of the VSYNC flip, reaction time is simply:

```
key, rtMs, _ := exp.ShowAndGetRT(target, responseKeys, 3000)
```

Internally `ShowAndGetRT` calls `exp.Keyboard.Clear()`, then `exp.ShowTS` to get the VSYNC flip timestamp, then `GetKeyEventTS`, and returns the difference in milliseconds. When you need the raw nanosecond timestamps (e.g. for EEG trigger alignment), you can call the primitives directly:

```
onset, _ := exp.ShowTS(target)    // nanoseconds at VSYNC flip
key, eventTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, 3000)
rtNS := int64(eventTS - onset)    // nanoseconds; divide by 1e6 for ms
```

This form is also the only correct approach when multiple stimuli are presented in sequence and you need RT relative to a specific one:

```
onset1, _ := exp.ShowTS(prime)    // prime appears
exp.Wait(500)
exp.ShowTS(target)                // target appears 500 ms later
key, eventTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, 3000)
rtToPrime := int64(eventTS - onset1) // RT from prime onset
```

### **Collecting multiple simultaneous responses — GetKeyEventsTS:**

In rare cases you may want to capture *all* key events, not just the first. `GetKeyEventsTS` is designed for detecting bilateral responses (e.g. both hands pressing at once) and works in two phases:

1. **Phase 1** — blocks until the first matching key arrives, respecting the timeout exactly like `GetKeyEventTS`.
2. **Phase 2** — waits up to 50 ms for any additional matching keys.



The second phase is necessary because human “simultaneous” presses are rarely truly simultaneous — the two KEY\_DOWN events typically arrive 10–50 ms apart. Without this window, a non-blocking drain after the first key would miss the second key almost every time.

```
events, err := exp.Keyboard.GetKeyEventsTS(responseKeys, 3000)
if len(events) > 0 {
    firstKey := events[0].Key
    firstTS  := events[0].TimestampNS
    rtNS     := int64(firstTS - onset)
}
if len(events) == 2 {
    lagNS := int64(events[1].TimestampNS - events[0].TimestampNS)
}
```

In the common single-response case only one event is returned, at the cost of at most 50 ms of extra latency before the function returns. For the typical single-response trial, GetKeyEventTS avoids that overhead entirely.

### Recording all presses over a fixed duration — CollectKeyEventsTS:

When the goal is to record *everything* the participant presses within a known time window (finger-tapping, free-response periods, stimulus-stream logging), use CollectKeyEventsTS. Unlike GetKeyEventsTS, it always runs for the full duration — it does not return early on the first keypress:

```
// Record every tap of F or J over 5 seconds
exp.Keyboard.Clear()
onset, _ := exp.ShowTS(stim)
events, err := exp.Keyboard.CollectKeyEventsTS(
    []control.Keycode{control.K_F, control.K_J}, 5000)
for _, ev := range events {
    rtNS := int64(ev.TimestampNS - onset)
    fmt.Printf("key %s at %d ms\n", ev.Key.KeyName(), rtNS/1_000_000)
}
```

Pass keys = nil to capture any key. Pass durationMS = 0 to do a non-blocking drain of whatever is already in the queue. Returns an empty (non-nil) slice if nothing was pressed.

### Querying whether a key is currently held — IsPressed:

IsPressed queries SDL’s internal scancode state array and returns true if the key is physically depressed at the instant of the call, regardless of the event queue. It is useful in animation loops that need to react continuously to a held key, or to verify that the participant has released a key before starting a new trial:

```
for exp.Keyboard.IsPressed(control.K_SPACE) {
    // do something while SPACE is held
    time.Sleep(10 * time.Millisecond)
}
```

Because `IsPressed` does not consume queue events, it can be called freely alongside `GetKeyEventTS` without interfering with the event stream.

### Measuring keypress duration — `WaitKeyReleaseTS`:

`WaitKeyReleaseTS` blocks until a `KEY_UP` event arrives for the specified key and returns its SDL3 hardware timestamp in nanoseconds. Together with the `KEY_DOWN` timestamp from `GetKeyEventTS`, this gives nanosecond-precision keypress duration:

```
key, downTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, -1)
// ... optionally update display while key is held ...
upTS, _ := exp.Keyboard.WaitKeyReleaseTS(key, 5000)
durationNS := upTS - downTS           // nanoseconds
durationMS := durationNS / 1_000_000 // milliseconds
```

This is the recommended approach for paradigms where press duration is a dependent variable (e.g., force-choice hold-to-respond, finger-tapping, or hold-triggered responses).

### Mouse

```
// Block until any button
btn, err := exp.Mouse.WaitPress()

// RT in milliseconds from call site (mirrors Keyboard.WaitKeysRT)
btn, rt, err := exp.Mouse.WaitPressRT(3000)

// Hardware-precision version: returns the SDL event's nanosecond timestamp
btn, eventTS, err := exp.Mouse.GetPressEventTS(3000)

// Non-blocking poll
btn, err := exp.Mouse.Check()

// Query whether a button is physically held down right now
held := exp.Mouse.IsPressed(sdl.BUTTON_LEFT)

// Wait for MOUSE_BUTTON_UP; returns its SDL hardware timestamp (nanoseconds)
upTS, err := exp.Mouse.WaitButtonReleaseTS(btn, 5000)

// Current cursor position in center-based coordinates
x, y := exp.Mouse.Position()

// Show / hide cursor
exp.Mouse.ShowCursor(false)
```

Button values: `sdl.BUTTON_LEFT`, `sdl.BUTTON_MIDDLE`, `sdl.BUTTON_RIGHT`, `sdl.BUTTON_X1`, `sdl.BUTTON_X2`.

`IsPressed` and `WaitButtonReleaseTS` mirror the keyboard's `IsPressed` and `WaitKeyReleaseTS` and can be used to measure click-hold duration:

```

btn, downTS, _ := exp.Mouse.GetPressEventTS(-1)
upTS, _       := exp.Mouse.WaitButtonReleaseTS(btn, 5000)
durationMS    := int64(upTS-downTS) / 1_000_000

```

## Multi-device input — WaitAnyEventTS

If you want to accept a response from *either* the keyboard or the mouse (or both), use `exp.WaitAnyEventTS` instead of calling `Keyboard` and `Mouse` separately:

```

onset, _ := exp.ShowTS(stim)

// Accept F or J key, or any mouse button click, up to 3 s
ev, err := exp.WaitAnyEventTS(
    []control.Keycode{control.K_F, control.K_J},
    true, // catchMouse
    3000,
)

rtNS := int64(ev.TimestampNS - onset) // hardware-precision RT, nanoseconds
rtMs := rtNS / 1_000_000

switch ev.Device {
case apparatus.DeviceKeyboard:
    fmt.Printf("key %d, RT %d ms\n", ev.Key, rtMs)
case apparatus.DeviceMouse:
    fmt.Printf("mouse button %d, RT %d ms\n", ev.Button, rtMs)
}

```

Pass `keys = nil` to accept any key. Pass `catchMouse = false` to ignore the mouse. On timeout, returns a zero `InputEvent` with `nil` error.

## Key codes

Key codes are re-exported in `control` so experiment code never needs to import `go-sdl3` just to name a key. Coverage includes the full alphabet (`K_A` ... `K_Z`), the digit row (`K_0` ... `K_9`), the numeric keypad (`K_KP_0` ... `K_KP_9`, `K_KP_ENTER`, `K_KP_PLUS`, `K_KP_MINUS`), navigation/control keys (`K_SPACE`, `K_ESCAPE`, `K_RETURN`, `K_BACKSPACE`, `K_TAB`, the arrow keys), and common punctuation (`K_MINUS`, `K_PLUS`, `K_EQUALS`, `K_LEFTBRACKET`, `K_RIGHTBRACKET`):

```

key, _ := exp.Keyboard.Wait()
if key == control.K_A { ... }

```

For a rarely-used code not in `defaults.go`, fall back to `go-sdl3/sdl` directly (e.g. `sdl.K_SEMICOLON`).

## Response Devices

All of the input methods above are device-specific: `GetKeyEventTS` for keyboards, `GetPressEventTS` for the mouse, and so on. If the experiment should run equally well with different input hardware — a keyboard in one lab, a response box in another — device-specific calls scatter conditional logic throughout the trial loop.

`ResponseDevice` is a single interface that abstracts over all of them:

```
// in package io
type ResponseDevice interface {
    WaitResponse(ctx context.Context) (Response, error)
    DrainResponses(ctx context.Context) error
    Close() error
}
```

`WaitResponse` blocks until a response is detected and returns a `Response`:

```
type Response struct {
    Source apparatus.DeviceKind // which device fired
    Code   uint32                 // key code, button index, or TTL bitmask
    RT     time.Duration           // elapsed from WaitResponse call
    Precise bool                   // timing quality (see below)
}
```

Wrap any input device with the matching adapter:

```
// Keyboard (SDL hardware timestamps → Precise: true)
var rd apparatus.ResponseDevice = &apparatus.KeyboardResponseDevice{KB: exp.Keyboard}

// Mouse (SDL hardware timestamps → Precise: true)
var rd apparatus.ResponseDevice = &apparatus.MouseResponseDevice{M: exp.Mouse}

// TTL response box, e.g. MEGTTLBox (software poll → Precise: false)
box, _ := triggers.NewMEGTTLBox("/dev/ttyACM0")
var rd apparatus.ResponseDevice = apparatus.NewTTLResponseDevice(box, 5*time.Millisecond)
```

All three look identical inside a trial loop:

```
onset, _ := exp.ShowTS(stim)
_ = rd.DrainResponses(ctx) // clear stale presses from previous trial
resp, err := rd.WaitResponse(ctx)
if err != nil { /* ESC, timeout, etc. */ }

rtMs := resp.RT.Milliseconds() // always valid
```

**Timing precision and the `Precise` flag** The `Precise` field tells you whether `RT` came from a nanosecond hardware event timestamp or a software poll:

| Device                   | Precise | RT accuracy                                                                                                   |
|--------------------------|---------|---------------------------------------------------------------------------------------------------------------|
| Keyboard, Mouse, Gamepad | true    | SDL3 hardware event timestamp, nanosecond resolution — identical to using <code>GetKeyEventTS</code> directly |
| MEGTTLBox, DLPIO8        | false   | <code>time.Now()</code> at poll detection; accuracy bounded by poll interval (default 5 ms)                   |

When `Precise` is true, the RT in the `Response` is computed from `sdl.TicksNS()` captured at the `WaitResponse` call versus the SDL3 hardware event timestamp — the same nanosecond clock used by `Screen.FlipTS`. It is therefore suitable for stimulus-onset-locked RT:

```
onset, _ := exp.ShowTS(stim)
resp, _ := rd.WaitResponse(ctx)
if resp.Precise {
    // resp.RT was computed from SDL hardware timestamps: nanosecond accuracy
    rtFromOnset := time.Duration(int64(onset) + resp.RT.Nanoseconds()) // approximate
}
```

When `Precise` is false, RT is still a valid elapsed duration — it just has ~poll-interval jitter (5 ms by default). For typical behavioral RT tasks this is acceptable; for EEG/MEG experiments requiring sub-millisecond synchrony, use the TTL output path for stimulus marking and treat RT as an approximate measure or use a hardware response box with sub-ms timestamping.

## 8. Data Collection

### Output files

When `exp.End()` is called, two files are written to `~/goxpy_data/`:

| File                                                                                       | Contents                                                                                                             |
|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <code>&lt;expName&gt;_sub-&lt;NNN&gt;_date-&lt;YYYYMMDD&gt;-&lt;HHMMSS&gt;.csv</code>      | Pure CSV data rows — directly importable by Excel, R, or pandas                                                      |
| <code>&lt;expName&gt;_sub-&lt;NNN&gt;_date-&lt;YYYYMMDD&gt;-&lt;HHMMSS&gt;-info.txt</code> | Session metadata: start/end time, hostname, OS, framework version, display and audio configuration, participant info |

Both files open automatically on `Initialize()` and are flushed to disk when `exp.End()` is called. Any spaces in `<expName>` are replaced by `-` so the resulting filenames never contain whitespace.

## Choosing the output directory

By default, data files are written to `~/goxpy_data/` (the folder name is the `results.DataFileDirectory` constant, appended to the user's home directory). To write them elsewhere, call `exp.SetOutputDirectory()` **before** `Initialize()` opens the files:

```
exp := control.NewExperiment("My Experiment", control.Black, control.White, 32)
exp.SetOutputDirectory("/path/to/mydata") // must come before Initialize()
if err := exp.Initialize(); err != nil { log.Fatal(err) }
defer exp.End()
```

The directory is created if it does not exist.

Because `NewExperimentFromFlags` calls `Initialize()` for you, overriding the directory requires the lower-level `NewExperiment(...)` + `Initialize()` path shown above rather than `NewExperimentFromFlags`.

## Declaring column names

Call this once, before the trial loop:

```
exp.AddDataVariableNames([]string{"condition", "response", "rt_ms", "correct"})
```

**Subject ID is always prepended automatically** — do not include it in the name list.

## Adding rows

```
exp.Data.Add(condition, response, rt, correct)
```

Fields are written in the same order as the variable names. Any type that has a meaningful `fmt.Sprint` representation works: `int`, `float64`, `bool`, `string`. Fields containing commas or quotes are escaped per RFC 4180.

## Example CSV output

```
subject_id,condition,response,rt_ms,correct
3,"congruent","F",412,true
3,"incongruent","J",538,false
```

Numbers and booleans are unquoted; strings are always double-quoted.

## Saving mid-experiment

For long experiments it is good practice to call `exp.Data.Save()` after each block. This flushes both files to disk so that data up to that point is not lost if the experiment crashes.

## 9. Stimuli: Lifecycle and Preloading

### GPU textures are lazily allocated

Creating a stimulus (e.g., `stimuli.NewTextLine(...)`) is cheap — it allocates a Go struct and stores the parameters, but does no GPU work. The SDL texture is created on the **first call to Draw**. This means:

- You can safely create all your stimuli before the run loop.
- The first presentation of each stimulus will be slightly slower due to texture upload.

For timing-sensitive presentations, preload explicitly:

```
stimuli.PreloadVisualOnScreen(exp.Screen, myStim)
// or, for a slice:
stimuli.PreloadAllVisual(exp.Screen, []stimuli.VisualStimulus{stim1, stim2, stim3})
```

Call this after `exp.Run` starts (the renderer is ready), but before the critical section:

```
err := exp.Run(func() error {
    // Preload everything while showing instructions
    exp.ShowInstructions("Loading, please wait...")
    stimuli.PreloadAllVisual(exp.Screen, stimSlice)

    // Now timing-sensitive trials
    for _, stim := range stimSlice { ... }
    return control.EndLoop
})
```

### Releasing textures

If you generate many stimuli dynamically (e.g., hundreds of unique text strings), call `stim.Unload()` after each trial to free the GPU memory:

```
for _, word := range wordList {
    stim := stimuli.NewTextLine(word, 0, 0, control.White)
    exp.Show(stim)
    exp.Keyboard.Wait()
    stim.Unload() // release GPU texture
}
```

For a fixed, small set of stimuli created once and reused throughout the experiment, you never need to call `Unload` — `exp.End()` handles cleanup.

### Writing a custom stimulus

Any type that implements `VisualStimulus` can be passed to `exp.Show`:

```
type VisualStimulus interface {
    Present(screen *apparatus.Screen, clear, update bool) error
    Preload() error
}
```

```

Unload() error
Draw(screen *apparatus.Screen) error
GetPosition() sdl.FPoint
SetPosition(pos sdl.FPoint)
}

```

Embed `stimuli.BaseVisual` to get the position methods and no-op Preload/Unload for free, then implement only Draw and Present:

```

type MyStimulus struct {
    stimuli.BaseVisual
    // your fields
}

func (m *MyStimulus) Draw(screen *apparatus.Screen) error {
    // draw using screen.Renderer SDL calls
    return nil
}

func (m *MyStimulus) Present(screen *apparatus.Screen, clear, update bool) error {
    return stimuli.PresentDrawable(m, screen, clear, update)
}

```

## Interactive widgets

Two stimuli present a UI and collect a structured response from the participant in a blocking loop. Neither is timing-critical, so they use `sdl.WaitEvent` rather than `VSYNC` polling.

**Menu** — a numbered, keyboard-navigable list:

```

m := stimuli.NewMenu([]string{"Beginner", "Expert", "Practice run"})
// Optional customisation:
m.HighlightColor = sdl.Color{R: 255, G: 220, B: 0, A: 255}

idx, err := m.Get(exp.Screen, exp.Keyboard, 0)
// idx is 0-based; -1 + sdl.EndLoop on ESC/quit

```

The selected item is shown with a > prefix in `HighlightColor`; all others use `TextColor`. Navigation: UP/DOWN arrows move the highlight; ENTER or SPACE confirms; digit keys 1–9 (0 for tenth) select directly without requiring confirmation.

**ChoiceGrid** — a grid of labelled buttons, activated by mouse click or matching key:

```

cg := stimuli.NewChoiceGrid([]string{"B", "C", "D", "F", "G"}, 3, "Pick 3 letters:")
selections, err := cg.Get(exp.Screen, exp.Keyboard)
// selections is []string in press order

```



## 10. High-Precision Streams (RSVP)

For paradigms that present stimuli at high speed — RSVP, attentional blink, priming — the standard `exp.Show / exp.Wait` cycle is not precise enough. The stream functions provide frame-accurate presentation:

- GC is **disabled** for the duration of the stream.
- Every onset and offset is aligned to a **VSYNC boundary**.
- A `TimingLog` records the predicted and actual onset for each stimulus.
- All keyboard and mouse events that occur during the stream are captured.

### Text stream

The simplest entry point:

```
words := []string{"CHAIR", "RIVER", "TIGER", "CLOCK", "STONE"}
on    := 150 * time.Millisecond
off   := 50 * time.Millisecond

events, logs, err := stimuli.PresentStreamOfText(
    exp.Screen, words, on, off,
    0, 0,           // center of screen
    control.White,
)
```

### Image / mixed stimulus stream

For images or any `VisualStimulus`:

```
stims := []stimuli.VisualStimulus{pic1, fixation, pic2, fixation}

// Regular cadence (all stimuli share the same on/off duration)
elements := stimuli.MakeRegularVisualStream(stims, 100*time.Millisecond, 0)

// Irregular cadence (individual onset times and durations, in ms)
elements, err := stimuli.MakeVisualStream(stims, onsetMs, durationMs)

events, logs, err := stimuli.PresentStreamOfImages(exp.Screen, elements, 0, 0)
```

### Reading events from the stream

Each `UserEvent` carries two timestamps:

- `Timestamp` — a `time.Duration` relative to stream start, Go monotonic clock, millisecond precision. Useful for quick inspection.
- `TimestampNS` — the SDL3 hardware event timestamp in nanoseconds, same clock as `Screen.FlipTS`. Use this for sub-millisecond RT computation.

```
for _, ev := range events {
    if ev.Event.Type == sdl.EVENT_KEY_DOWN {
```

```

        key := ev.Event.KeyboardEvent().Key
        t    := ev.Timestamp.Milliseconds() // ms from stream start (coarse)
        fmt.Printf("key %d pressed at %d ms\n", key, t)
    }
}

```

Use `stimuli.FirstKeyPress` to find the first matching key-down event without writing a loop:

```

if ev, ok := stimuli.FirstKeyPress(events, sdl.K_SPACE); ok {
    fmt.Printf("Space pressed at %d ms\n", ev.Timestamp.Milliseconds())
}

```

Both `ev.Timestamp` (Go clock, ms precision) and `ev.TimestampNS` (SDL3 hardware, nanoseconds) are available on the returned `UserEvent`.

## Computing RT from stream events

Because `UserEvent.TimestampNS` and `TimingLog.OnsetNS` are on the same SDL nanosecond clock, reaction time from a specific stimulus is exact:

```

for _, ev := range events {
    if ev.Event.Type == sdl.EVENT_KEY_DOWN {
        for _, l := range logs {
            if ev.TimestampNS >= l.OnsetNS && (l.OffsetNS == 0 || ev.TimestampNS < l.OffsetNS) {
                rtNS := int64(ev.TimestampNS - l.OnsetNS)
                fmt.Printf("RT from stimulus %d: %d ms\n", l.Index, rtNS/1_000_000)
            }
        }
    }
}

```

## Reading the timing log

```

for i, l := range logs {
    jitter := (l.ActualOnset - l.TargetOn).Milliseconds()
    fmt.Printf("stimulus %d: target %d ms, actual %d ms, jitter %d ms\n",
        i, l.TargetOn.Milliseconds(), l.ActualOnset.Milliseconds(), jitter)
}

```

`OnsetNS` and `OffsetNS` give the SDL3 nanosecond timestamps of the actual VSYNC flips that turned each stimulus on and off. These are zero for stimuli that had zero frames in their on or off period.

Jitter below  $\pm 1$  frame ( $\pm 8$ – $17$  ms depending on monitor) is normal and expected. Larger jitter indicates system load or GPU driver issues.

## Platform timing notes

PresentStreamOfImages provides the best timing accuracy that the operating system and display driver allow. The characteristics differ by platform:

### Linux

- Without a compositing window manager (e.g. a plain X11 session with no compositor), SDL3's `Present()` blocks until the next VSYNC boundary. Onset jitter is typically **< 1 ms**.
- With a Wayland compositor or an X11 compositing WM (KWin, Mutter, Picom), the compositor controls buffer swaps. Onset jitter is typically **1-3 ms**; the compositor may add one frame (~17 ms at 60 Hz) of fixed latency.
- For the most reliable timing on Linux, disable the compositor or use a plain X11 session.

### macOS (Metal)

- The macOS WindowServer compositor is **always active** — exclusive fullscreen does not bypass it. SDL3 submits each frame to the compositor, which forwards it to the display at the next VSYNC.
- `FlipTS` captures `sdl.TicksNS()` immediately after `Present()` returns; this reflects when the frame was *submitted to the compositor*, not when photons reached the screen. The additional compositor latency is typically **1-2 frames** (~17-33 ms at 60 Hz).

### Windows

- In **exclusive fullscreen** mode (`fullscreen=true`), the DWM (Desktop Window Manager) compositor is bypassed. Timing behaviour is similar to Linux without a compositor — jitter is typically **< 1 ms**.
- In **windowed mode**, DWM is always active and adds approximately one frame of compositor latency. Jitter is typically **1-3 ms**.
- For the most reliable timing on Windows, always run in fullscreen mode.

### What OnsetNS measures

`TimingLog.OnsetNS` is recorded immediately after `screen.FlipTS()` returns, not at the moment photons reach the screen. On top of the compositor latency noted above, there is additional hardware pipeline latency (GPU scan-out, cable propagation, display panel response) of typically **0-2 frames**. This latency is constant across trials and does not affect within-experiment RT precision, but it must be accounted for if absolute (photodiode-verified) onset times are required.

### Minimum duration

Because presentation is VSYNC-locked, durations shorter than one frame period (e.g. < 16.7 ms at 60 Hz) are rounded up to the nearest whole frame. A stimulus requested for 50 ms on a 60 Hz display is shown for exactly **3 frames (50.0 ms)**; one requested for 60 ms is shown for **4 frames (66.7 ms)**.

## Audio stream

The same model applies to sounds:

```
// All tones must be pre-loaded before the stream
for _, t := range tones {
    t.PreloadDevice(exp.AudioDevice)
}

elements := stimuli.MakeRegularSoundStream(tones, 200*time.Millisecond, 100*time.Millisecond)
events, logs, err := stimuli.PlayStreamOfSounds(elements)
```

## Mixed stream

PresentStreamOfStimuli presents a single **sequential** stream that mixes visual and audio stimulus types. It reuses the VSYNC-locked visual loop, so visual elements are frame-accurate; audio elements are triggered once at the slot's first flip and the previous frame is **held** (not cleared) for the slot — place a visual just before a sound to keep it visible while the sound plays.

```
tone.PreloadDevice(exp.AudioDevice) // audio elements must be device-bound first

elements := []stimuli.StreamElement{
    {Stimulus: stimuli.NewTextLine("Ready?", 0, 0, control.White), DurationOn: 600 * time.Millisecond},
    {Stimulus: pic, DurationOn: 800 * time.Millisecond}, // stays on screen during the sound
    {Stimulus: tone, DurationOn: 400 * time.Millisecond, DurationOff: 200 * time.Millisecond},
}

events, logs, err := stimuli.PresentStreamOfStimuli(exp.Screen, elements, 0, 0)
```

Concurrent audio-visual overlap (a sound and an animating visual at the same instant) is not yet supported; the stream is strictly one slot at a time.

---

## 11. Audio

### Sounds from files or embedded bytes

```
// From a file
snd := stimuli.NewSound("ping.wav")
snd.PreloadDevice(exp.AudioDevice)
snd.Play()
snd.Wait() // block until playback finishes

// From embedded bytes (go:embed)
//go:embed assets/beep.wav
var beepWav []byte

snd := stimuli.NewSoundFromMemory(beepWav)
```

```
snd.PreloadDevice(exp.AudioDevice)
snd.Play()
```

## Procedural tones

```
tone := stimuli.NewTone(1000.0, 200*time.Millisecond, 200) // 1 kHz, 200 ms, medium vol
tone.PreloadDevice(exp.AudioDevice)
tone.Play()
```

Volume is 0–255. Use `PreloadDevice` once; then call `Play` repeatedly.

## Segment playback with fade

Play only part of a longer sound, with smooth fade-in and fade-out:

```
snd.PlaySegment(1.5, 3.0, 0.05) // play 1.5–3.0 s, 50 ms ramps
```

This is useful for stimuli like vowels extracted from longer recordings.

## Built-in feedback sounds

```
exp.Audio.PlayBuzzer() // play the embedded error/incorrect sound asynchronously
exp.Audio.PlayCorrect() // play the embedded correct/reward sound asynchronously
```

Both return immediately; the sound plays in the background.

## Synchronous vs asynchronous

`snd.Play()` starts playback and returns immediately. `snd.Wait()` blocks until it finishes. For trial timing where you need to know when a sound ended, call both:

```
snd.Play()
doSomethingElse()
snd.Wait() // wait here if needed
```

For fire-and-forget feedback sounds, call `snd.Play()` without `snd.Wait()`.

## Microphone recording

`goxpyriment` can capture audio from the system microphone via SDL3's recording API. The primary use cases are:

- **Picture naming** — measure the latency from image onset to the participant's first vocalization.
- **Vocal shadowing** — measure the latency between a spoken or tonal stimulus and the participant's repetition of it.

Open the microphone once, before the trial loop:

```
// nil uses the default recording spec: F32LE mono 44100 Hz
if err := exp.OpenMicrophone(nil); err != nil {
    log.Fatalf("microphone unavailable: %v", err)
}
// exp.Microphone is now ready; it is closed automatically by exp.End()
```

To select a custom format (e.g. higher sample rate or stereo):

```
spec := &sdl.AudioSpec{Format: sdl.AUDIO_F32LE, Channels: 1, Freq: 48000}
exp.OpenMicrophone(spec)
```

To list all connected recording devices and open a specific one:

```
devices, _ := apparatus.GetRecordingDevices()
for _, d := range devices {
    name, _ := d.Name()
    fmt.Println(d, name)
}
mic, _ := apparatus.NewMicrophoneFromDevice(devices[0], nil)
exp.Microphone = mic
```

## Voice key — detecting vocal onset

A VoiceKey monitors the microphone and fires when the amplitude exceeds a threshold. It is the goxpyriment equivalent of a hardware voice key or PsychoPy's OnsetVoiceKey.

```
vk := apparatus.NewVoiceKey(exp.Microphone, 0.03, 128)
// threshold = 0.03 (RMS amplitude 0–1 for F32LE)
// windowSz = 128 samples ≈ 2.9 ms at 44100 Hz
```

## Per-trial pattern:

```
// 1. Arm: flush the mic buffer and start capturing.
// Returns the SDL3 nanosecond timestamp of capture start.
captureStartNS, _ := vk.Arm()

// 2. Present the stimulus.
imageOnsetNS, _ := exp.ShowTS(picture) // picture naming
// – or –
stimulusNS, _ := tone.PlaySyncedWithFlip(exp.Screen) // shadowing

// 3. Wait for the vocal response (2-second window).
onsetNS, pcm, err := vk.WaitOnset(captureStartNS, 2000)
if errors.Is(err, apparatus.ErrVoiceTimeout) {
    // participant did not respond in time
}

// 4. Compute RT (both timestamps are on the SDL3 nanosecond clock).
rtMs := int64(onsetNS - imageOnsetNS) / 1_000_000
```

WaitOnset also returns pcm — the raw PCM bytes captured during the trial. Save them as WAV for offline verification (see below).

**Threshold calibration.** A value of 0.02–0.05 works in a quiet room. If you observe:

- **Spuriously short RTs (< 100 ms)** — the threshold is too low; it is triggering on breath noise, lip smacks, or microphone bleed from the speakers. Raise the threshold or use headphones.
- **detected = false (missed responses)** — the threshold is too high for soft or breathy voices. Lower it.

Always verify using the saved WAV files rather than adjusting the threshold blindly.

### Timing: microphone and screen on the same clock

Both imageOnsetNS (from exp.ShowTS or screen.FlipTS) and the voice onset computed by WaitOnset use `sdl.TicksNS()` — SDL3's monotonic nanosecond clock. No cross-clock conversion is needed:

```
vk.Arm() ————— captureStartNS (sdl.TicksNS())
           |
           | exp.ShowTS(picture)
           | ↳ VSYNC flip ————— imageOnsetNS (sdl.TicksNS())
           |
           | [participant starts speaking at sample N]
           |
           |———— onsetNS = captureStartNS + N × 1e9 / sampleRate
                               (sdl.TicksNS() domain)
```

$RT = (onsetNS - imageOnsetNS) / 1\,000\,000 \text{ [ms]}$

Call `vk.Arm()` just before showing the stimulus so the microphone is already capturing when the image appears. There is no need to arm it long in advance.

### Saving WAV files for verification

Always save a WAV file for each trial and spot-check a random subset after the session. Voice keys can fire on breaths, lip smacks, or room noise; the WAV is the ground truth.

```
apparatus.WriteWAV(
    fmt.Sprintf("sub-%03d_trial-%02d.wav", exp.SubjectID, trial),
    exp.Microphone.Spec,
    pcm,
)
```

WriteWAV supports F32LE, S16LE, S32LE, and U8 formats and writes a standard RIFF/WAV file that any audio editor can open.

For post-hoc re-analysis of saved files (e.g. applying a different threshold), use `apparatus.ScanOnset`:

```
sampleIndex := apparatus.ScanOnset(pcm, newThreshold, 128)
if sampleIndex >= 0 {
    revisedOnsetNS = apparatus.SampleOnsetNS(captureStartNS, sampleIndex, int(exp.Micro
})
```

### Acoustic feedback (shadowing and AV tasks)

When audio plays through laptop built-in speakers, the microphone picks up the playback signal and the voice key triggers on it rather than on the participant's voice. This produces spuriously short latencies (sometimes < 0 ms if the mic captures the speaker before the amp threshold is crossed).

**Always use headphones for playback** in any task that simultaneously records and plays audio.

If headphones are unavailable, a simple analysis-side fix is to reject any onset that falls within a guard interval after stimulus onset (e.g. discard `onsetNS - stimulusNS < 100 ms`).

---

## 12. Experimental Design and Randomization

### When to use `design.Block` / `design.Trial`

The design package provides a structured Experiment → Block → Trial hierarchy with string-keyed factors. Use it when:

- You have a multi-block experiment and want `exp.Design.Blocks` to drive the loop.
- You want between-subjects Latin-square counterbalancing (`AddBWSFactor`).

For simple single-block experiments, a plain Go slice of a custom struct is often more readable (and type-safe). Both approaches are valid.

### Building a design

```
block := design.NewBlock("main")
for _, cond := range []string{"congruent", "incongruent"} {
    t := design.NewTrial()
    t.SetFactor("condition", cond)
    t.SetFactor("soa", 200)
    block.AddTrial(t, 20, false) // 20 copies, appended in order
}
block.ShuffleTrials()
exp.AddBlock(block, 1)
```

Iterate at runtime:

```
for _, blk := range exp.Design.Blocks {
    for _, trial := range blk.Trials {
```



```

    cond := trial.GetFactor("condition").(string)
    soa := trial.GetFactor("soa").(int)
    // ...
}
}

```

## Randomization helpers

The design package provides randomization functions that work on any slice type:

```

// In-place shuffle (generic – works on any []T)
design.ShuffleList(mySlice)

// Random integer in [a, b] inclusive
design.RandInt(500, 1500)

// Random element from a slice
word := design.RandElement(wordList)

// Weighted coin flip
if design.CoinFlip(0.75) { /* 75% chance */ }

// Truncated normal sample in [a, b]
design.RandNorm(800.0, 1200.0)

// Shuffled integer range [first, last]
order := design.RandIntSequence(0, len(stimuli)-1)

```

## Between-subjects counterbalancing

```

// Register once during setup
exp.AddBWSFactor("mapping", []interface{}{"F=left/J=right", "F=right/J=left"})

// At runtime – returns the condition assigned to this subject's ID
mapping := exp.GetPermutedBWSFactorCondition("mapping").(string)

```

The assignment follows a balanced Latin square so that conditions rotate across subjects (subject 1 → condition A, subject 2 → condition B, subject 3 → condition A, ...).

## Constrained shuffling

For designs where you need to prevent the same condition from appearing more than N times in a row, use block-level `AddTrial` with `randomPosition: true`, which inserts each trial at a random position rather than appending:

```

for _, cond := range conditions {
    t := design.NewTrial()
    t.SetFactor("condition", cond)
}

```

```
block.AddTrial(t, 5, true) // 5 copies each, randomly interleaved during construct
}
```

---

## 13. Embedding Stimuli and Trial Lists in the Executable

One of Go's most useful deployment features is `//go:embed`: a compiler directive that copies files from disk into the binary at build time. The result is a **single self-contained executable** that carries all its images, sounds, and trial-list CSV files with it — no separate assets/ folder, no installer, no missing-file errors on lab computers.

**When to embed.** Embedding is the right choice for stimuli that are fixed at design time: images, sounds, and pre-generated trial lists. It is not appropriate for stimuli you generate procedurally at runtime (textures drawn with Canvas, tones created with NewTone, text rendered from NewTextLine), or for data files that are written during the experiment.

### 13.1 Project layout

Collect all static assets in a subdirectory alongside `main.go`:

```
myexperiment/
  main.go
  assets/
    stim_A.png
    stim_B.png
    feedback_correct.wav
    trials.csv
```

### 13.2 Embedding individual image and sound files

Import `_ "embed"` (blank identifier, needed to activate the directive) and declare a `[]byte` variable per file:

```
package main

import _ "embed"

//go:embed assets/stim_A.png
var stimAImg []byte

//go:embed assets/stim_B.png
var stimBImg []byte

//go:embed assets/feedback_correct.wav
var feedbackWav []byte
```

Pass the byte slices to the `FromMemory` constructors:

```
import "github.com/chrplr/goxpyriment/stimuli"

stimA := stimuli.NewPictureFromMemory(stimAImg, 0, 0)
stimB := stimuli.NewPictureFromMemory(stimBImg, 0, 0)
feedback := stimuli.NewSoundFromMemory(feedbackWav)
```

These constructors behave identically to `NewPicture` and `NewSound`; textures are still allocated lazily on first `Draw`, and `PreloadAllVisual` / `PreloadDevice` work the same way.

### 13.3 Embedding a whole stimulus directory

When you have many images (e.g. one per trial) it is impractical to declare a variable for each one. Embed the whole directory as an `embed.FS` and iterate over it at runtime:

```
package main

import "embed"

//go:embed assets/stimuli
var stimuliFS embed.FS
```

Load every PNG in the directory into a map:

```
import (
    "fmt"
    "github.com/chrplr/goxpyriment/stimuli"
)

pics := map[string]*stimuli.Picture{}

entries, err := stimuliFS.ReadDir("assets/stimuli")
if err != nil {
    log.Fatal(err)
}
for _, e := range entries {
    if e.IsDir() {
        continue
    }
    data, err := stimuliFS.ReadFile("assets/stimuli/" + e.Name())
    if err != nil {
        log.Fatal(err)
    }
    pics[e.Name()] = stimuli.NewPictureFromMemory(data, 0, 0)
}

// Use by filename:
exp.Show(pics["face_happy.png"])
```

Or load a specific file by name:

```
data, _ := stimuliFS.ReadFile(fmt.Sprintf("assets/stimuli/face_%s.png", condition))
stim := stimuli.NewPictureFromMemory(data, 0, 0)
```

### 13.4 Embedding a CSV trial list

A CSV file that defines trial order and parameters is a natural fit for embedding. The workflow is: embed → read into []byte → wrap in bytes.NewReader → parse with encoding/csv.

**Example assets/trials.csv:**

```
condition,soa_ms,correct_key
congruent,200,F
incongruent,200,J
congruent,400,F
incongruent,400,J
```

**Embedding and parsing:**

```
package main

import (
    "bytes"
    _ "embed"
    "encoding/csv"
    "log"
    "strconv"
)

//go:embed assets/trials.csv
var trialsCSV []byte

type trial struct {
    condition string
    soaMs      int
    correctKey string
}

func loadTrials() ([]trial, error) {
    r := csv.NewReader(bytes.NewReader(trialsCSV))
    records, err := r.ReadAll()
    if err != nil {
        return nil, err
    }
    var trials []trial
    for i, rec := range records {
        if i == 0 {
            continue // skip header row
        }
    }
}
```

```

    }
    soa, _ := strconv.Atoi(rec[1])
    trials = append(trials, trial{
        condition: rec[0],
        soaMs:      soa,
        correctKey: rec[2],
    })
}
return trials, nil
}

```

For **tab-separated** files, set `r.Comma = '\t'` immediately after creating the reader.

For **semicolon-separated** files (common in European locales), set `r.Comma = ';'.`

Use the loaded trials in the experiment loop:

```

trials, err := loadTrials()
if err != nil {
    log.Fatalf("loading trials: %v", err)
}

err = exp.Run(func() error {
    exp.ShowInstructions("Press F or J to respond.")
    for _, t := range trials {
        exp.Blank(500)
        stim := stimuli.NewTextLine(t.condition, 0, 0, control.White)
        key, rt, _ := exp.ShowAndGetRT(stim, []control.Keycode{control.K_F, control.K_J})
        correct := string(key.KeyName()) == t.correctKey
        exp.Data.Add(t.condition, t.soaMs, rt, correct)
    }
    return control.EndLoop
})

```

### 13.5 Mixing embedded and runtime-generated stimuli

Embedding is selective — embed only what is fixed, generate the rest at runtime:

```

//go:embed assets/mask.png
var maskImg []byte

// Fixed stimulus: loaded from embedded bytes
mask := stimuli.NewPictureFromMemory(maskImg, 0, 0)

// Dynamic stimulus: generated from trial parameters
target := stimuli.NewTextLine(word, 0, 0, control.White)

```

## 13.6 Build and deployment

Nothing changes in the build command:

```
go build -o myexperiment .
```

The assets/ directory must exist on disk at build time but is **not needed at runtime**. Copy the single binary to any lab computer running the same OS:

```
# Cross-compile for Windows from Linux
GOOS=windows GOARCH=amd64 go build -o myexperiment.exe .
```

```
# Cross-compile for macOS (Apple Silicon)
GOOS=darwin GOARCH=arm64 go build -o myexperiment-mac .
```

The embedded assets are included in all cross-compiled outputs automatically.

**Binary size.** Each embedded file adds its uncompressed size to the binary. A typical experiment with a few dozen PNG images and WAV files will add 5–50 MB — acceptable for a self-contained research tool. For very large stimulus sets (hundreds of high-resolution images, video files), consider distributing them as a separate archive alongside the binary instead of embedding.

---

## 14. Animated Stimuli

Three functions run self-contained VSYNC-locked animation loops. All three:

- Disable GC for the duration of the loop.
- Drain the SDL event queue before the first frame.
- Return a MotionResult with the response key, mouse button, and reaction time.

```
type MotionResult struct {
    Key      sdl.Keycode // interrupt key pressed (0 if none)
    Button   uint8      // mouse button (0 if none)
    RTms     int64      // ms from first frame to response; total elapsed on timeout
}
```

### Moving dot cloud

```
result, err := stimuli.PresentMovingDotCloud(
    exp.Screen,
    100,                // number of dots
    3.0,                // dot radius (px)
    150.0,              // cloud radius (px)
    control.Origin(),   // center at (0, 0)
    150.0,              // speed in px/sec
    5000,               // max duration ms (0 = infinite)
    []control.Keycode{control.K_SPACE}, // interrupt keys
```

```

    false,                // catch mouse?
    control.White,        // dot color
    control.Color{A: 0},  // background (transparent)
)
fmt.Printf("RT: %d ms\n", result.RTms)

```

## Drifting grating

```

result, err := stimuli.PresentMovingGrating(
    exp.Screen,
    400, 400,           // width, height (px)
    control.Origin(),   // center
    0.0,               // orientation (degrees; 0 = vertical bars drifting right)
    0.05,              // spatial frequency (cycles per pixel)
    2.0,               // temporal frequency (Hz)
    0.8,               // contrast [0, 1]
    0.5,               // mean luminance [0, 1]
    3000,              // max duration ms
    []control.Keycode{control.K_SPACE},
    false,
)

```

## Drifting Gabor patch

```

result, err := stimuli.PresentMovingGabor(
    exp.Screen,
    200,                // bounding box size (px); at least 6×sigma
    30.0,               // sigma (Gaussian SD in px)
    control.Origin(),   // center
    45.0,               // orientation
    0.05,               // spatial frequency (cycles/px)
    3.0,               // temporal frequency (Hz)
    0.9,               // contrast
    0.5,               // mean luminance
    0,                  // max duration (0 = until response)
    []control.Keycode{control.K_SPACE},
    false,
)

```

---

## 15. Gamma Correction and Luminance Linearity

### The gamma problem

Standard monitors apply a power-law transfer function when converting digital RGB values to physical luminance:

$$L(V) = k \cdot (V/255)^\gamma \quad \gamma \approx 2.2 \text{ for sRGB/LCD}$$

This means equal steps in RGB values do **not** produce equal steps in physical luminance (cd/m<sup>2</sup>). For example, with  $\gamma = 2.2$ :

| RGB value | Physical luminance (% of max) |
|-----------|-------------------------------|
| 0         | 0%                            |
| 64        | 4.9%                          |
| 128       | 21.6%                         |
| 192       | 52.2%                         |
| 255       | 100%                          |

This matters in psychophysics experiments where luminance values are specified as physical quantities (e.g. contrast masks, threshold estimation, magnitude estimation).

### Inverse-gamma LUT

goxpyriment corrects for this by pre-computing a 256-entry look-up table (LUT) per channel. For each desired linear luminance value  $v$  (0–255), the LUT stores the digital value required to achieve it:

$$\text{LUT}[v] = 255 \cdot (v/255)^{(1/\gamma)}$$

Applying the LUT before sending a color to the renderer means the monitor's forward gamma converts it back to the intended linear value.

### Enabling gamma correction

```
// After exp.Initialize() and before the trial loop:
exp.SetGamma(2.2) // typical sRGB monitor

// Or with per-channel calibration (from photometer measurements):
exp.GammaCorrector = apparatus.NewGammaCorrector(2.1, 2.2, 2.3)
```

Then wrap every color with `exp.CorrectColor`:

```
// Specify colors in linear luminance space (0–255).
// exp.CorrectColor maps them to the physical digital values.
disk := stimuli.NewFilledCircle(exp.CorrectColor(control.RGB(128, 128, 128)), radius)
```

When `SetGamma` has not been called, `CorrectColor` is a no-op that returns the color unchanged, so it is safe to always call it.

### Measuring your monitor's gamma

The most accurate approach is photometer-based: measure luminance at several RGB levels, fit the model  $L(V) = k \cdot (V/255)^\gamma$ , and pass the resulting  $\gamma$  to `SetGamma`. Without a photometer,  $\gamma = 2.2$  is a reasonable default for modern sRGB displays.



## Practical example

The examples/Magnitude-Estimation-Luminosity example accepts a `-gamma` flag:

```
go run main.go -gamma 2.2
```

With the flag set, the 7 luminance levels (10, 25, 50, 100, 150, 200, 255) are treated as linear targets and corrected before display, so they produce physically uniform luminance steps.

---

## 16. Hardware Triggers and TTL Devices

EEG and MEG recordings require a synchronisation signal — a short TTL pulse sent at the exact moment a stimulus is presented — so that electrophysiological data can be time-locked to experimental events. The `triggers` package provides this, along with the ability to read TTL inputs from response hardware such as fiber-optic response pads.

### Concepts

All trigger devices in `goxpyriment` share two interfaces:

- **OutputTTLDevice** — send trigger codes (set lines HIGH/LOW, generate pulses)
- **InputTTLDevice** — read response button states (poll or block)

Lines are **0-indexed (0-7)**. Bit N of a bitmask drives line N.

### Sending triggers (EEG event codes)

```
import (
    "github.com/chrplr/goxpyriment/triggers"
    "time"
)

// Auto-detect a DLP-I08-G; falls back to NullOutputTTLDevice (no-op) if absent.
out, _, err := triggers.AutoDetectDLPI08()
if err != nil { log.Fatal(err) }
defer out.Close()

// In the trial loop:
onset, _ := exp.ShowTS(stim)
out.Pulse(0, 10*time.Millisecond) // 10 ms pulse on line 0 = EEG event marker
```

For the NeuroSpin MEGTTLBox:

```
box, err := triggers.NewMEGTTLBox("/dev/ttyACM0")
defer box.Close()
```

```
box.Pulse(0, 5*time.Millisecond) // single line
box.PulseMask(0b00000011, 5*time.Millisecond) // lines 0 and 1 simultaneously
```

## Timing advice

The output pulse should be sent as close as possible to `exp.ShowTS(stim)`. Because both `Screen.FlipTS` and the trigger output use the system's real-time clock, the latency between the VSYNC flip and the TTL edge is typically under 1 ms. The EEG amplifier records this edge, and the exact onset can be recovered by subtracting `int64(triggerEdgeNS - onsetNS)` if the amplifier's sample clock is synchronised.

## Reading response inputs (FORP pads)

The MEGTTLBox wires a fiber-optic response pad (fORP) to its 8 TTL input lines. Use `WaitForInput` directly or via the `ResponseDevice` wrapper (section 7):

```
// Via InputTTLDevice directly
_ = box.DrainInputs(ctx) // clear stale presses
mask, rt, err := box.WaitForInput(ctx)
buttons := triggers.DecodeMask(mask)
for _, b := range buttons {
    fmt.Println(b) // e.g. "left blue", "right red"
}

// Via ResponseDevice
rd := apparatus.NewTTLResponseDevice(box, 5*time.Millisecond)
_ = rd.DrainResponses(ctx)
resp, _ := rd.WaitResponse(ctx)
// resp.Code == bitmask, resp.Precise == false (software poll)
```

**fORP button mapping** (NeuroSpin wiring):

| FORPButton constant | Line | Arduino pin | STI channel |
|---------------------|------|-------------|-------------|
| FORPLeftBlue        | 0    | D22         | STI007      |
| FORPLeftYellow      | 1    | D23         | STI008      |
| FORPLeftGreen       | 2    | D24         | STI009      |
| FORPLeftRed         | 3    | D25         | STI010      |
| FORPRightBlue       | 4    | D26         | STI012      |
| FORPRightYellow     | 5    | D27         | STI013      |
| FORPRightGreen      | 6    | D28         | STI014      |
| FORPRightRed        | 7    | D29         | STI015      |

## FT232H (Adafruit FT232H breakout, Linux)

A USB-to-MPSSE bridge. AD0-AD7 are TTL output lines; AC0-AC7 are TTL input lines. No `libftdi` or `D2XX` installation needed — the driver uses Linux `usbfs` directly.

```

box, err := triggers.NewFT232H()
defer box.Close()

box.Pulse(0, 5*time.Millisecond)
mask, rt, _ := box.WaitForInput(ctx)

// Auto-detect (returns NullOutputTTLDevice if no device found):
out, _ := triggers.AutoDetectFT232H()

```

Prerequisites: unload the ftdi\_sio kernel module (sudo rmmod ftdi\_sio) and ensure the user has rw access to the USB device node (udev rule or plugdev group membership).

### GPIO (Raspberry Pi and other Linux SBCs)

Drives any 8 GPIO pins as TTL outputs and reads any other 8 pins as TTL inputs, using the Linux GPIO character device v2 API. Works on Raspberry Pi (BCM pin numbers), Rock Pi, BeagleBone, Jetson, and any Linux SBC with kernel  $\geq$  5.10.

```

box, err := triggers.NewLinuxGPIOTrigger(
    triggers.WithGPIOOutputPins([8]int{17, 27, 22, 5, 6, 13, 19, 26}),
    triggers.WithGPIOInputPins([8]int{12, 16, 20, 21, 4, 25, 24, 23}),
)
defer box.Close()

box.Pulse(0, 5*time.Millisecond)
mask, rt, _ := box.WaitForInput(ctx)

```

Output-only and input-only configurations are both valid. Prerequisites: kernel  $\geq$  5.10; user in the gpio group.

### LabJack T4 (Modbus TCP)

Communicates over Ethernet via Modbus TCP — no LabJack SDK or driver required. FIO0–FIO7 are output lines; EIO0–EIO7 are input lines.

```

box, err := triggers.NewLabJackT4("192.168.1.100")
defer box.Close()

box.Pulse(0, 5*time.Millisecond)
mask, rt, _ := box.WaitForInput(ctx)

```

The device must be reachable on the local network. Default Modbus port 502 is used; append :port to the host string to override.

### Supported devices

| Device               | Type                          | Output | Input | Connection            | Notes                                          |
|----------------------|-------------------------------|--------|-------|-----------------------|------------------------------------------------|
| DLP-IO8-G            | DLPI08                        | ✓      | ✓     | USB-CDC serial        | ASCII protocol                                 |
| NeuroSpin MEGTTL-Box | MEGTTLBox                     | ✓      | ✓     | USB serial            | Arduino Mega; binary protocol; fORP input      |
| Adafruit FT232H      | FT232HTrigger                 |        | ✓     | USB (usbfs)           | Linux only; no libftdi needed                  |
| Linux GPIO           | LinuxGPIOTrigger              |        | ✓     | GPIO pins             | Any Linux SBC; kernel $\geq$ 5.10              |
| LabJack T4           | LabJackT4                     | ✓      | ✓     | Ethernet (Modbus TCP) | No SDK needed                                  |
| LPT parallel port    | ParallelPort                  |        | —     | LPT                   | Linux only; ppdev ioctl                        |
| None / fallback      | NullOutputTrigger (Developer) |        | —     | —                     | Returned by AutoDetectDLPI08, AutoDetectFT232H |

## 17. Display Compositor Bypass

A compositing window manager (compositor) intercepts every application frame and blends it with other on-screen elements before sending the result to the display. This introduces additional latency and jitter that are problematic for millisecond-accurate stimulus presentation. The degree to which goxpyriment can avoid this overhead depends on the operating system.

On **Linux with X11**, SDL3 automatically sets the `_NET_WM_BYPASS_COMPOSITOR` window property when the application enters fullscreen mode. Compliant compositors (KWin, Mutter, Compton) respond by *unredirecting* the fullscreen window — routing its framebuffer directly to the display scan-out pipeline without compositing. The result is presentation latency below 1 ms, comparable to a compositor-free session. On **Linux with Wayland**, the compositor is always the intermediary for frame delivery, but modern compositors (KWin 5.21+, GNOME Mutter 44+) support *direct scan-out* for fullscreen applications, achieving similar low-latency behavior in practice. The best achievable timing on Linux is obtained by running without any display

server: setting the environment variable `SDL_VIDEODRIVER=kmsdrm` before launching the experiment directs SDL3 to communicate with the Linux KMS/DRM subsystem directly, bypassing X11 and Wayland entirely. This configuration — typically used from a virtual terminal (`Ctrl+Alt+F2`) — is recommended for the most demanding timing requirements such as single-frame subliminal stimulation or high-frequency RSVP.

On **Windows**, SDL3's fullscreen mode creates an exclusive fullscreen surface that bypasses the Desktop Window Manager (DWM), again yielding frame-interval jitter typically below 0.3 ms (one standard deviation), comparable to Linux.

**macOS** is the exception. Apple's Metal rendering pipeline always delivers frames to the display through the WindowServer compositor, and no third-party application can bypass it — even in fullscreen.

These platform differences can be quantified empirically with the display and frames sub-tests of the bundled Timing-Tests suite.

---

## 18. Variable Refresh Rate (VRR) Stimulus Presentation

Standard displays refresh at a fixed rate (e.g., 60 Hz), which quantises every stimulus duration to multiples of the frame period (16.67 ms at 60 Hz). A researcher wishing to present a stimulus for 10 ms, 17 ms, or 23 ms cannot do so on such a display without relying on temporal dithering or hardware frame-blending techniques that introduce their own artefacts. Variable Refresh Rate (VRR) technology — marketed as AMD FreeSync, NVIDIA G-Sync, or the VESA Adaptive-Sync standard — removes this constraint by allowing the display to hold each frame for an arbitrary duration and refresh only when instructed to do so by the GPU.

`goxpyriment` exposes VRR through a single API call: `exp.Screen.SetVSync(0)` disables the VSYNC block, so that every subsequent `SDL_RenderPresent` call returns immediately. The following pattern then achieves arbitrary stimulus durations with duration error typically below 0.5 ms (one standard deviation):

1. Draw the stimulus and call `screen.FlipTS()`, which presents the frame and records `onsetNS` (an SDL nanosecond timestamp) immediately after `Present()` returns.
2. Busy-wait for the desired duration  $d$ : spin until `time.Now() >= onset + d`. The last  $\sim 500$   $\mu$ s use a CPU spin loop (already used in the trigger timing routines) to eliminate OS scheduling jitter.
3. Draw the blank screen, call `FlipTS` again to record `offsetNS`.

The actual software-controlled duration is  $(\text{offsetNS} - \text{onsetNS}) / 1e6$  ms. On a VRR-capable display this equals the true on-screen duration plus a small, constant hardware pipeline latency (GPU scan-out and panel response, typically 1–5 ms) that can be calibrated once with a photodiode using the frames timing sub-test of the Timing-Tests suite.

Every VRR panel supports VRR only within a finite refresh-rate window (e.g., 48–144 Hz, corresponding to frame durations of 6.9–20.8 ms for a typical gaming monitor).

Durations outside this window cause the panel to revert to fixed-rate behaviour, re-introducing quantisation. The Timing-Tests suite includes a dedicated vrr sub-test that sweeps target durations from 1 ms to a user-specified maximum in 1 ms steps and reports duration errors at each step, directly revealing the panel's VRR window in the output data.

On a display without VRR support, disabling VSYNC produces frame tearing. The vrr sub-test detects this automatically: on a non-VRR display the duration errors cluster at multiples of the frame period rather than being uniformly small, providing an unambiguous diagnostic.

---

## 19. Putting It All Together

Here is a skeleton that illustrates how the concepts compose in a realistic experiment:

```
package main

import (
    "fmt"
    "log"

    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/design"
    "github.com/chrplr/goxpyriment/stimuli"
)

const (
    NReps          = 10
    FixationDuration = 500
    ResponseTimeout = 3000
)

type trialDef struct {
    word  string
    color control.Color
    name  string
}

func main() {
    exp := control.NewExperimentFromFlags("Color Word Task", control.Black, control.White)
    defer exp.End()

    exp.AddDataVariableNames([]string{"trial", "word", "ink", "key", "rt_ms", "correct"})

    // Build trial list
    words := []string{"RED", "GREEN", "BLUE"}
    colors := []control.Color{control.Red, control.Green, control.Blue}
```

```

names := []string{"red", "green", "blue"}
keys  := []control.Keycode{control.K_R, control.K_G, control.K_B}

var trials []trialDef
for _, word := range words {
    for j := range colors {
        trials = append(trials, trialDef{word, colors[j], names[j]})
    }
}
for i := 0; i < NReps-1; i++ {
    trials = append(trials, trials[:9]...)
}
design.ShuffleList(trials)

// Preload stimuli
fixation := stimuli.NewFixCross(20, 2, control.White)

err := exp.Run(func() error {
    // Preload fixation cross (it will be used every trial)
    stimuli.PreloadVisualOnScreen(exp.Screen, fixation)

    exp.ShowInstructions(
        "Name the INK COLOR of each word.\n\n" +
        "R = Red   G = Green   B = Blue\n\n" +
        "Press SPACE to start.",
    )

    for i, t := range trials {
        // Fixation
        exp.ShowTimed(fixation, FixationDuration)

        // Stimulus + Response
        stim := stimuli.NewTextLine(t.word, 0, 0, t.color)
        key, rt, _ := exp.ShowAndGetRT(stim, keys, ResponseTimeout)

        // Score
        correct := false
        for j, k := range keys {
            if key == k && names[j] == t.name {
                correct = true
                break
            }
        }

        exp.Data.Add(i, t.word, t.name, key, rt, correct)
        fmt.Printf("trial %3d  %s/%s  rt=%dms  %v\n", i, t.word, t.name, rt, correct)
    }
})

```

```

        if !correct {
            exp.Audio.PlayBuzzer()
        }

        // Release texture (new TextLine each trial)
        stim.Unload()

        exp.Blank(500)
    }

    return control.EndLoop
})

if err != nil && !control.IsEndLoop(err) {
    log.Fatalf("experiment error: %v", err)
}
}

```

Key patterns illustrated:

- NewExperimentFromFlags + defer exp.End() at the top.
- exp.AddDataVariableNames before the run loop.
- design.ShuffleList for trial randomization.
- stimuli.PreloadVisualOnScreen inside exp.Run for timing-sensitive stimuli.
- exp.Show for single stimuli; exp.Blank for ITIs.
- exp.Keyboard.WaitKeysRT for response + RT.
- stim.Unload() for dynamically created stimuli.
- exp.Audio.PlayBuzzer() for feedback.
- return control.EndLoop at the end.

---

*For the complete function signatures and type definitions, see the [API Reference](#). For more worked examples, browse the [examples/](#) directory.*